



Challenges for efficient land use in rice production of northern Iran: The use of modern cultivars among small-scale farmers

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ABSTRACT

In recent decades, precision technologies and agricultural innovations have spread widely in rice production, but a lag in adoption, particularly in developing countries, is often noted. A survey of 400 rice farmers was conducted on the basis of a structured questionnaire to collect data on the nature and reasoning behind the use of improved rice cultivars by paddy farmers in Guilan Province, Iran. Improved cultivars were used by 42.8% of the farmers. Almost half of the farmers perceived high profitability (55.2%) and high importance (49.8%) of this technology. Nevertheless, a noticeable fraction of the farmers perceived no profitability (14.8%) or no importance (12.5%) of this technology, while 9.8% and 16.2% of the farmers were not aware about these features of the technology. A close cooperation with extension agents about common production problems was observed for most farmers (60.5%), but most farmers had low cooperation with extension agents (58.5%) and low participation in activities recommended by the local agricultural offices (59.8%). Logistic regression analysis revealed that the adoption of improved cultivars was determined by the perceived profitability of the technology, the perceived importance (necessity) of the technology, the experience in rice growing, and the number of livestock, with a positive relationship. Findings provide opportunities for more efficient land use in rice production by promoting the use of modern rice cultivars among small-scale farmers. Low adoption of improved cultivars is not attributed to the inappropriateness of the technology, but rather with the inconsistency of the technology with farmers' socio-economic conditions. Any attempt to promote adoption should consider the role of farmers in the information transfer to other farmers. Providing subsidized long-term facilities to low-income farmers' groups would also promote adoption.

1. Introduction

The recent decades have witnessed a growing interest in new farming technologies for rice production. In this context, the adoption of alternative farming practices by farmers is an attempt to exploit the new technologies and accomplish ecological and profitable ways of production using the physical, chemical, and biological resources (Bigdeli and Pedigi, 2010; Villano et al., 2015). Adoption of new technologies is expected to provide farmers with opportunities for boosting rice production and improving incomes (Mariano et al., 2012; Askary Bozayeh et al., 2017), but the dissemination of new technologies among rural communities will be successful, if they can change knowledge, attitudes, and behavior of farmers, so that the new technologies can be ultimately adopted. One major reason for the limited adoption of agricultural technologies is often farmers' unawareness of

the agricultural technologies and the lack of appropriate promotion of innovations (Kalantari and Khadem Adam, 1997; Simtowe et al., 2016). Potential remedies lie in the promotion of agricultural innovations that improve agricultural productivity and thus increase the efficiency of smallholder agriculture.

Guilan Province enjoys unique attributes for the agricultural sector owing to its moderate climate and fertile lands for crop cultivation. Among various agricultural activities, the province has a dominant role in rice production that has long been known as a strategic commodity in the area (Keshavarz et al., 2010; Golichenari et al., 2014a, b). A large part of the rice fields in Iran is still under local varieties, because of their excellent quality traits (Rani, 1998). However, low yields (averaging 2.5 to 3.5 tonnes per ha) as well as lodging and susceptibility to blast (*Pyricularia grisea*) and stem borer (*Chilo* spp.) limit production. One way to enhance return per unit area is to use new technologies

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Table 1
Most common improved varieties in paddy fields of Guilan Province.

Variety	Aroma	Kernel length (mm)	Amylose content (%)	Paddy yield (kg/ha)	Resistance level	
					Rice Blast	Stem borer
Khazar	None	7.39	24.9	5000-6000	R	MR
Sepeed rood	None	7.28	27.2	6500-7000	R	S
Kadous	None	8.06	23.8	5500-6000	R	R
Gohar	Strong	8.20	23.8	8500-9000	MR	-
Gil 1	Strong	7.10	28.0	4000-4500	R	-
Gil 3	Strong	7.50	26.0	5000-5500	R	-
Bejar	Medium	6.00	25.1	6500-7000	R	-

R: resistant; MR: moderately resistant; S: susceptible.

(Dinpanah et al., 2009; Khonje et al., 2015). The use of appropriate technologies, in addition to generating significant growth in agricultural production, may reduce production costs and entail economical savings. After the establishment of Rice Research Institute of Iran in 1993, the main target was related to research activities for the development of modern rice varieties that produce high yield and are resistant to diseases and pests. The most common improved rice cultivars used in paddy fields of Guilan Province along with basic characteristics are illustrated in Table 1. However, farmers are often unwilling to adopt new technologies because various environmental, organizational, and socio-economic factors place barriers against adoption (Mariano et al., 2012).

The analysis of technology adoption is a very common topic in the literature. For example, in many Asian countries, the advent of the Green Revolution compelled many researchers to focus on the acceptance of agricultural innovations (Mariano et al., 2012). Education, experience, and the relationship with agricultural extension services affected positively the use of modern rice varieties in Nepal (Joshi and Pandey, 2005). Analyzing the adoption of modern rice varieties in southwestern Nigeria showed that land ownership, contact with extension agents, and cropping area had a significant positive effect on the adoption of the improved cultivars, while there was not a significant difference between adopters and non-adopters in terms of gender, type of agronomic system (e.g., monocropping or intercropping), and membership in associations (Saka et al., 2015). Education, land size, and experience in maize growing positively and significantly affected the adoption of hybrid maize varieties in Punjab, Pakistan (Iqbal et al., 1999). However, the age of farmers did not have a significant effect on hybrid maize varieties' adoption. The adoption of high-yielding cultivars was negatively impacted by uncertainty in the price of seed at the time of sale, low quality, and low availability of high-yielding varieties, low profitability in comparison with local cultivars, lack of timely supply of inputs, long cropping period, high water and fertilizer requirements, and susceptibility to pests and diseases, which acted as inhibitors (Keshavarz et al., 2010). Seed availability and absence of seed policy were four main checks in rice production increase in Tillabéry, Niger (Koussoré et al., 2015). A recent study in Ghana recommended increasing farmers' education, ensuring access to extension services, and improving varietal characteristics for increasing farmers' adoption of improved maize varieties (Alhassan et al., 2016).

Due to the saturation of the cropping area and the constraints on the expansion of rice farm lands in Guilan Province, the attempts to increase rice production should focus on methods that can increase yield per unit area as the main choice in the agenda. Among the various methods of yield improvement, the use of modern inputs, including improved varieties, can be very promising. Nevertheless, adoption rates are always challenged by the design of the transfer process to the target community, which is characterized by its own features. Therefore, the determination of factors contributing to adoption of improved cultivars among farmers is necessary for the implementation of policy control measures to improve livelihoods through sustainable rice production.

The aim of the current study was to explore what drives farmers to use of improved cultivars in the eastern part of Guilan Province. Findings are expected to shed some light on how technology could be promoted among farmers and may have important implications for seed delivery systems in the study area.

2. Materials and methods

2.1. Study site

The study was conducted in Rudsar County of Guilan Province in northern Iran. Rudsar County is located in the eastern part of Guilan Province. The county has an area of 1,354 km², of which 18,878 ha are devoted to agricultural land. An area of 7,053 ha is cultivated with paddy rice, producing 26,224 tonnes of rice. The statistical population of this county consisted of all farmers living in Rudsar County, including farmers who adopted the technology of improved varieties and farmers who did not.

2.2. Sample selection

To determine sample size required for the study, the least sample size table needed for a definite community with categorical data provided by Bartlett et al. (2001) was used. Accordingly, the sample size of the research was determined to be 370 people. To avoid errors due to questionnaire non-return, an additional 10% of the determined number of participants was included, resulting in 400 farmers as the final sample size of the study.

2.3. Survey tool and data collection

After a comprehensive review of literature (Joshi and Pandey, 2005; Keshavarz et al., 2010; Sow et al., 2015) a questionnaire was developed as the main tool of the research. The questionnaire included the dependent variable of the adoption or non-adoption of improved cultivars (adopter = 1 and non-adopter = 0) and the independent variables of personal, economic, social, and technical-agricultural attributes. Social cooperation of farmers was evaluated on a five-point Likert-type scale as: very low, low, moderate (fair), high, and very high. Social participation can be broadly defined as a person's involvement in activities that provide interaction with other individuals. Also, the profitability and importance of improved cultivars as well as the usefulness of information resources were evaluated on a five-point Likert-type scale from none to very high. Profitability refers to how much the technology can boost farmers' financial interests, while importance refers to the usefulness and extent of necessity of the technology to the farmers. The content (face) validity of the questionnaire was determined by faculty members of the Agricultural Management Department of Mohaghegh Ardabili University, professors of agricultural extension and agronomy in Islamic Azad University of Rasht, and experts of Jihad-e-Agriculture Organization. To determine the reliability of the questionnaire, a

preliminary test was conducted, in which 30 questionnaires were administered to participants outside the sample. The total Cronbach's alpha was estimated at 0.91, indicating the high reliability of the research tool. Data were collected with face-to-face interviews with the farmers. A simple random sampling strategy was chosen in such a way that every individual of the target population had an equal chance to be selected in the sample. For this purpose, farmers' lists were obtained from the local authorities and participants were randomly selected using the numbered lists.

2.4. Data analysis

Frequencies, percentages, means, and standard deviations were calculated for demographic features of the respondents and the independent and dependent variables, concerning descriptive statistics. Also, logistic regression analysis was used to capture the role of each variable in the adoption or non-adoption of the improved cultivars technology. The reason for the selection of logistic regression analysis was the fact that the dependent variable was of dichotomous type (yes, no). This method calculates the probability of each level of the dichotomous variable based on the independent variables. Thus, the relationship of using improved cultivars with each independent variable, including profitability of the technology, necessity (importance) of the technology, farmers' social participation, along with demographic, economic, and technical variables were determined, so that the impact of each independent variable on the dependent variable can be explained. The SPSS software package was used for data analysis.

3. Results

3.1. Farmers' socio-economic background

The mean age of the respondents was 51.26 years, with the youngest and the oldest participants being 25 and 80 years old, respectively (Table 2). The highest frequency (45.5%) was in the class of 40–50 years. According to Table 2, 31.5% of the respondents had academic degrees, 26.2% had a high school diploma, and 42.3% had elementary or intermediate education. Only a small portion of the farmers (2.3%) were illiterate. The experience in rice growing varied in the range of 5 to 65 years with an average of 29.73 years. The highest frequency of experience was in the category of 20 to 30 years (34.2%).

The surveyed rice farmers had at least no livestock and at most 35 heads of livestock, with an average of 4.23 heads (Table 3). The class of no livestock had the highest frequency in livestock number. The respondents had, on average, 4.46 poultry, but the highest frequency of rice farmers was in the class of no poultry (36.0%) and 1 to 20 poultry. The minimum agricultural income was reported to be 10

Table 2
Frequency distribution of demographic features (n = 400).

Variable	Frequency	Percentage	Mean	SD
Age (years)				
< 40	58	14.5	51.26	10.872
From 40 to 50	182	45.5		
From 51 to 60	78	19.5		
≥ 60	82	20.5		
Education level				
Illiterate	9	2.3	-	-
Elementary and intermediate schools	160	40.0		
High school and diploma	105	26.2		
Academic education	126	31.5		
Experience in rice growing (years)				
< 20	131	32.8	29.73	14.815
From 20 to 30	137	34.2		
From 31 to 40	44	11.0		
≥ 40	88	22.0		

Table 3
Economic features of the rice farmers (n = 400).

Feature	Frequency	Percentage	Mean	SD
Number of livestock				4.23 7.289
0	240	60.0		
1–10	98	24.5		
> 10	62	15.5		
Number of poultry			24.46	53.478
0	144	36.0		
From 1 to 20	128	32.0		
From 21 to 40	76	19.0		
> 40	52	13.0		
Agricultural income (million IRR)			117557500	105674165.56
< 50	159	39.8		
From 50 to 100	108	27.0		
> 100	133	33.2		
Non-agricultural income (million IRR)			91350000	103998397.42
0	7	1.8		
< 50	116	29.0		
From 50 to 100	110	27.5		
> 100	90	22.5		
Rice production (kg)			2153.25	1483.236
< 1500	119	29.8		
From 1500 to 3000	110	27.5		
> 3000	90	22.5		
Total agricultural expense (million IRR)			48632500	52430284.96
< 2	116	29.0		
From 2 to 30	93	23.2		
> 30	191	47.8		

million IRR (Iranian rials) and the maximum 500 million IRR, with an average of 117,557,500 IRR. However, the highest frequency of rice farmers (39.8%) was in the class of < 50 million IRR. The non-agricultural income ranged from 10 to 500 million IRR, with an average of 91,350,000 IRR. However, the highest frequency of rice farmers (30.5%) was in the class of < 50 million IRR. The maximum and minimum rice production rate ranged between 300 and 12,000 kg, with an average 2,153.25 kg. The agricultural expenses of farmers were at least 2 million IRR and at most 280 million IRR, with an average of 48,632,500 IRR and the highest frequency in the class of > 30 million IRR. The farmers had at least 0.1 ha and at most 10 ha of farming land. Mean farm area was 1.60 ha, but the highest frequency of rice farmers (67.2%) was in the range of < 1 ha (Table 4). The respondents had at least 4.96 land parcels. The highest frequency (53.0%) was related to > 3 farm parcels. Mean distance from farm was 2.10 km, with the highest distance being 15 km. However, the highest frequency of rice farmers (52.5%) was related to the range of < 1 km. In addition, mean distance of farmers' house from road was 0.92 km, with the highest frequency (56.2%) being in the range of 0.1 to 1.0 km.

The frequency of items pertaining to social participation of rice growers are shown in Table 5. Most farmers had high 'cooperation with other farmers', high 'cooperation with *Dehyari*' and high 'cooperation with rural entities and cooperatives', but low participation in programs recommended by Jihad-e Agriculture and low cooperation with extension agents. Cooperation with agricultural extension agents was ranked the last.

3.2. Extent of use of improved cultivars technology

A large proportion of the farmers (42.8%) adopted the technology of improved cultivars, but most farmers did not (Table 6). Table 7 summarizes the extent of profitability and importance of improved cultivars

Table 4
Technical features of the rice farmers surveyed (n = 400).

Features	Frequency	Percentage	Mean	SD
Land size (ha)				
< 1	269	67.2	1.60	1.802
From 1 to 2	68	17.0		
> 2	63	15.8		
Number of farm parcels				
1	106	26.5	4.96	4.186
2	56	14.0		
3	26	6.5		
> 3	212	53.0		
House distance to farm (km)				
< 1	210	52.5	2.10	2.333
From 1 to 3	120	30.0		
> 3	70	17.5		
House distance to road (km)				
< 0.1	93	23.2	0.92	0.731
From 0.1 to 1	225	56.2		
> 1	82	20.5		

Table 5
Social participation among rice farmers of Rudсар county (n = 400).

Social participation	VL	L	M	H	VH
	Percentage (%)				
Cooperation with other farmers about production problems	2.0	8.5	29.0	28.5	32.0
Cooperation with <i>Dehyari</i> and Islamic Council of village	10.0	7.0	28.5	30.0	24.5
Membership and cooperation with rural entities and cooperatives	8.3	19.0	21.2	35.0	16.5
Attendance in educational-promotional courses	18.5	18.0	34.0	23.8	
Cooperation in programs recommended by Jihad-e Agriculture	36.2	23.6	12.2	19.5	8.5
Cooperation with extension agents	51.5	7.0	20.0	15.3	6.2

VL: very low, L: low, M: moderate, H: high, VH: very high

Table 6
Adoption of high-yielding rice cultivars among rice farmers of Rudсар county (n = 400).

High-yielding rice cultivars	Frequency	Percentage
Non-adoption	229	57.2
Adoption	171	42.8

Table 7
Profitability and importance of high-yielding rice cultivars to rice farmers (n = 400).

High-yielding rice cultivars	Do not know	None	Low	High	Very high
Profitability	9.8	14.8	20.2	41.2	14.0
Importance	16.2	12.5	21.5	36.0	13.8

Table 8
Usefulness of information sources among farmers.

Information source	Mean	SD
Farmers	2.95	0.974
TV & Radio	2.51	1.014
Extension agent	1.60	1.494
Educational movie	1.59	1.374
Research center	1.28	1.360
Internet	1.14	1.405

Table 9
Effective factors underpinning the adoption of improved rice cultivars in Rudсар County.

Variables	B	S.E.	Wald	P-value		Exp(B)
Profitability	1.154	0.25	21.50	0.000	**	3.170
Importance	1.08	0.22	24.06	0.000	**	2.940
Experience in rice growing	0.06	0.01	26.06	0.000	**	1.060
Number of livestock	0.05	0.02	5.86	0.015	*	1.050
Constant	-6.96	0.77	81.52	0.000	**	0.001

–2 log likelihood = 286.070; Cox & Snell R^2 = 0.406; Nagelkerke R^2 = 0.542.

** Significant at $P < 0.01$; * significant at $P < 0.05$.

Percentage of correct prediction = 81.0%.

technology to the farmers. Profitability refers to how much the technology can boost farmers' financial interests and importance refers to the usefulness and extent of necessity of the technology to the farmers. Almost half of the farmers (55.5%) stated that the technology of high-yield rice cultivars is highly profitable, while 41.5% perceived it to be of low profit. Similarly, 41.5% of the farmers (49.5%) felt that improved cultivars are of high importance, whereas 41.5% assessed this technology of low importance. The usefulness of information resources of farmers are shown in Table 8. Information from other farmers showed the highest usefulness (mean = 2.95).

3.3. Logistic regression analysis

In logistic regression, –2 log likelihood is the statistic that is extensively used as a measure of model's goodness of fit. This statistic acts like a standard where the estimated logit model matches well with data, its value is low. According to Table 9, this statistic was found to be 286.070 showing the model's high goodness of fit. Also, its percentage of correct prediction was 81.0%. This means that 81.0% of the observations were distinguished correctly. Thus, we can claim that the logistic model was capable of predicting a high percentage of dependent variable's values by the variables inserted in the model (including profitability, farmer's age, education level, experience in rice growing, number of livestock, number of poultry, agricultural income, non-agricultural income, rice production rate, agricultural expense, participation and cooperation, land size, number of land parcels, and house distance to road). Furthermore, the significance of χ^2 at $P < 0.01$ showed to what extent the factors determined by the model were effective in the adoption of improved seed technology by local rice growers.

The results revealed that the variables of profitability and importance of the technology, experience in rice growing, and number of livestock determined the adoption of high-yielding cultivars, with a positive relationship. Also, the Wald values presented in Table 9 indicated that the variables of experience in rice growing (26.06), importance of the technology (24.06), and profitability of the technology (21.50) had the highest relative share in the prediction of improved cultivars adoption by the studied farmers (Table 9). Exp(B) odds ratio values showed that the profitability of the technology had the largest contribution in predicting adoption, so that as profitability increases by one unit, the likelihood of the adoption of an improved cultivar is increased by 3.170 units, holding other factors constant.

4. Discussion

In this study, factors that determine why farmers choose to grow improved rice cultivars were explored in a sample of 400 rice farmers in Rudсар County of Guilan Province in northern Iran. Logistic regression analysis was used to find the relationship of using improved cultivars with several independent variables, such as the profitability and the necessity (importance) of the technology, along with socio-demographic, economic, and technical variables, so that the impact of each

independent variable on the dependent variable can be explained. Findings shed some light on how this technology could be promoted among farmers and may have important implications for seed delivery systems. This information can point to appropriate policies aimed at increasing adoption of improved cultivars among non-adopters that could reduce poverty levels and increase household food security.

The coefficients in the regression model showed a significant and positive relationship between the adoption of improved cultivars and the profitability of the technology ($P < 0.01$). This finding is consistent with those of previous studies (Nazemolsadat et al., 2006; Lashgarara and Asadi, 2008), indicating that the use of technologies can be encouraged by stating their advantages and benefits to rice growers or by their application at a limited scale to demonstrate their profitability. Perception of low profitability of improved seeds may be a reason for high variation in the spread of adoption (Amare et al., 2012; Minten et al., 2013). The use of stone terraces for soil and water conservation in Ethiopia was positively influenced by farmers' perceptions of profitability of the technology (Amsalu and De Graaff, 2007). Daberkow and McBride (2003) concluded that without informational constraints, the adoption decision for precision technology adoption in USA would only depend on profitability of the technology. However, if there are informational constraints then the adoption may also depend on farmers' awareness. Robertson et al. (2012) noted an uncertainty in the advantage of variable fertilization rate in Western Australia as a constraint undermining confidence for non-adopters of the technology. The main elements towards this direction are the suppression of cost for the technology and precise statements for the profitability of the technology. From this point of view, simplifying the technology to enable farmers recognize the monetary benefit is necessary. Moreover, in our study, the importance (necessity) of the technology influenced the adoption of improved cultivars positively and significantly ($P < 0.01$), indicating that the more tailored the technologies are with farmers' requirements, the higher the rate of their adoption. Previous studies noted that farmers' decision to adopt a technology relied on the importance and usefulness of the technology (García-Lambán et al., 2012; Abdollahzadeh et al., 2017). Therefore, it is imperative for research centers to identify rice growers' needs and develop appropriate technologies, so that farmers show more attention to them. Moreover, policies geared at promoting modern technologies in rice fields should point out their usefulness to farmers to ensure continuous use after initial adoption.

Rice growing experience had a favorable influence on the adoption of improved seed technology ($P < 0.01$). This is in line with previous studies (Joshi and Pandey, 2015; Idris et al., 2012; Onumadu and Osahon, 2014), which reported a positive impact of farming experience on the adoption of improved technologies. Similarly, lack of experience was noted as a main problem in the adoption of modern rice production technology in India (Srinivasan and Ponnarasi et al., 2009). Experience upgrades farmers' competence in production, implying a lower level of uncertainty about innovations' performance and also an ability to evaluate the advantage of the technology considered. Therefore, farming experience can be an indication of practical knowledge that farmers have acquired to deal with certain inherent farm production and adoption problems (Onumadu and Osahon, 2014). In the early stages of technology adoption, farming experience can be particularly important because the potential benefits of the technology are still tested. Thus, gradual development of the technology and continuous retraining of farmers are essential for promoting the adoption (Ainembabazi and Mugisha, 2014). This finding is expected because experienced farmers are more skillful and have better access to new information about improved technologies. It could also imply that knowledge gained over time may help farmers in evaluating information thereby influencing their adoption decision (Idris et al., 2012). Farmers are engaged in the generation of innovations, as they can impulse providers or co-developers, apart from being only adopters (Busse et al., 2014).

There was a significant and positive influence of number of livestock ($P < 0.05$) on the adoption of improved cultivars. In other words, as number of livestock grows, farmers exhibit more willingness to adopt the improved cultivars technology. So, during the introduction of new technologies to rice growers, especially at the first phases of introducing high-cost technologies, the presentations should target farmers with high number of livestock. Similarly, owning oxen significantly and positively affected the probability of adopting improved rice varieties by farm households (Ghimire et al., 2015). It appears that animal power that is available for plowing is a significant factor where the cropped area is small and perhaps tractor mechanization is unprofitable. Off-farm income can be used to purchase inputs for farming. In this regard, wealthy farmers are supposed to have sufficient support in terms of resources (e.g., off-farm income) that facilitate investment on new technologies (Nyariki, 2011). For example, livestock income is often considered as an indicator of the availability of working capital, which facilitates farming activities (Ponnamperuma, 2003). In this regard, livestock activities help in the diversification of income source of a household and, therefore, allow mobilization of other resources that can be used to improve farm productivity (Lansing and Mungoma, 2008). However, subsistence farmers should carefully assess the relevance of a technology to their income level and food security status, prior to decision whether to adopt or not. Thus, farmers with livestock income may be less willing to invest on a technology, if non-farm alternatives might be more rewarding to them (Thu et al., 2014). In this context, a negative effect of livestock on the adoption of stone terraces as a conservation practice for soil and water was observed in a watershed in Ethiopia, showing a tendency of households to focus more on livestock than on crop production (Amsalu and De Graaff, 2007).

According to the logistic regression analysis of this study, the adoption of improved cultivars was not influenced significantly by farmer's age, education level, number of poultrys, rice production rate, total expenses, land size, number of land parcels, and house distance to farm. However, other studies reported a negative relationship between participants' age and adoption of innovations in question (Ahmed et al., 2016; Arslan et al., 2014). On the other hand, a positive impact of age on adoption also has been observed (Abdollahi Ezzatabadi and Islamlouyian et al., 2007). An old farmer means more experience in production practices and greater physical capital, thereby facilitating innovation adoption. On the other hand, advanced age can also be linked with short-term planning, loss of energy, and risk averseness. In this regard, the effect of age on the adoption of a new technology can be considered indeterminate. With reference to other variables, adopters of sprinkler irrigation technology had more land parcels than non-adopters, indicating that land fragmentation was not a barrier in the adoption behavior towards that technology (Bagheri and Malekmohammadi, 2011). However, a negative impact of the number of farm parcels on innovation adoption has been reported (Kohansal et al., 2009). Moreover, high yields of some local varieties had a strong negative influence on the adoption of improved rice cultivars (Keshavarz et al., 2010).

5. Policy implications

Widespread use of advanced technologies and practices is critical for intensifying food production in an environmentally sustainable manner. This study provided useful information about factors that determine why farmers choose to grow improved rice cultivars in a sample of 400 rice farmers in Rudsar County of Guilan Province in northern Iran. This information can point to appropriate policies aimed at increasing adoption of improved cultivars among non-adopters that could reduce poverty levels and increase household food security.

First, it is imperative to explore educational methods underpinning the adoption of improved cultivars in rice production. Extension and advisory services are critical for facilitating smallholders' access to technology and knowledge. Robust extension systems and services

disseminate and communicate information to farmers through messages that are clear, tailored to the learning needs of target audiences, and locally relevant. Nevertheless, both theory and practice show that adoption requires going beyond simple awareness of a technology. Therefore, farmers must try technologies, make modifications to match specific on-farm circumstances, and receive support when troubleshooting or when questions arise. Policies geared at promoting modern technologies in rice fields should simplify the technology to enable farmers to recognize the monetary benefit and also point out the usefulness of the technology to farmers to ensure continuous use after initial adoption.

Second, since income directly impacts the adoption of improved cultivars and given the high risk of the fulfillment of this technology, it is recommended to provide low-income groups with subsidized, long-term facilities. Farmers may need additional incentives to try technologies and practices in the first place. Incentives that include a price subsidy scheme allowing farmers' groups, particularly young farmers, to purchase new seeds at lower than market price would promote adoption. With reference to seed delivery system, although communication and information flows may work well with direct relationship among actors, these flows may be weak in the absence of direct relationships. Therefore, increasing availability of certified seed, affordability, and seed packaging would promote the use of certified seed among farmers.

Third, any attempt to promote improved cultivars should consider the role of farmers in information transfer to other farmers. Diffusion of improved technologies among small-scale farmers can be enhanced through the participation of farmers. For example, farmers as trainers of other farmers could effectively disseminate fairly simple technologies, such as new crop varieties as compared to more complex ones with high risks, such as certain crop protection measures. This approach successfully diffuse new technologies from lead to group farmers and also to nearby farmers, but it remains a big challenge for diffusing technologies further to the wider community.

6. Conclusions

Factors that determine why farmers choose to grow improved varieties were explored in a sample of 40 rice farmers in Rudsar County of Guilan Province in northern Iran. The adoption of improved seeds was significantly influenced by perceived profitability and the perceived importance (necessity) of the technology, farmers' social participation, farming experience, agricultural income, non-agricultural income, and house distance to road. Appropriate policies aimed at increasing adoption of improved varieties that could reduce poverty levels and increase household security could be put forth according to findings. Low adoption is not totally related to the inappropriateness of the technology, but it is associated with the inconsistency of the technology with the farming socio-economic conditions. For example, families with educated members who are more aware of reduced crop productivity and need more information, adopt the technology of improved seeds more easily.

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